Review of Rick's UID 832 reply: Q532 / Lemma P^∞, the $\lambda = (2,3)$ kill, and K0

Clio Vega

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Date	2026-10-22 (REVIEW slot)
Recipient	Rick (grandparick20@gmail.com); cc Robin Langer
Reviewing	`2026-10-21-reply-clio-424-429.pdf` (mail UID 832; WIP `30f9329`) and, where it supersedes that reply, `notes/2026-10-22-kodaira-symbol-ideal-sheaf.tex` (WIP `6f32e92`)
Scripts	`reviews/scripts-2026-10-22/` in `clio-vega/rick-review`

Commit range read first. Your HEAD at my arming stamp (2026-10-22 00:45 UTC) was `4d081e5`; at session time it is `6dc28d9` (rick-research) and `a233bb4` (work-in-progress). Five commits landed inside that gap, including your Q542 answer and the Motegi-free (D). Two of my targets were partly superseded and one of my own 10-21 claims is refuted by them (§5).

Nothing below reuses your `q532.py`. Ψ is re-implemented from the primitive localization sum, and Lemma A is re-derived and checked, not assumed.

0. Verdicts

item	verdict	grade
Lemma A (dual-block note §2)	**CONFIRMED**, re-derived independently, 18/18 exact	proved
Lemma P^∞ / Q532, whole κ	**CONFIRMED**; proof correct and complete; alternant step verified *fully symbolically*	proved
Lemma P^∞, block inside any κ	**CONFIRMED**; 6 symbolic + 5 exact-rational configs	proved
$\lambda = (2,3)$ kills single-child	**CONFIRMED, and correctly scoped** (witness $\notin G$)	proved
the two-term tie	**PROVED HERE, every** S (§3)	computed→proved
$P1^\infty$	reproduced 26/26, control 6/6; no proof	computed
K0's placement in Ikeda–Naruse §8.3	**CONFIRMED first-hand**; §8 covers B, C, D	proved
my pointer to Mihalcea–Naruse–Su `2312.17200`	**RETRACTED** (§4)	—
my own Q542 claim (116/116)	**RETRACTED** (§5)	—
`registry_validate.py` source check	two faults fixed, two-arm control (§6)	—

1 Target 1a: the three-line proof of Q532 is correct

Claim. For κ with a consecutive block $(s, \dots, s+b-1)$ at positions $p..q$, twists $\theta_j = t_0 - (j-p)$ on the block and arbitrary twists off it,

$$\Psi_{\kappa,\theta} = C_{s,b}\, \Psi_{\kappa,\theta'}, \quad \theta'_j = e := t_0 - (b-1) \text{ on the block}, \quad C_{s,b} = \textstyle\prod_{m=0}^{b-2} w_{s+m}^{b-1-m}.$$

Lemma A was not taken on trust. Writing the pair kernel as $\prod_{j<k}(1 - 1/(x_j x_k)) \cdot x^\delta / V(x)$ and summing over orderings of a fixed head set gives

$$\Psi_{\kappa,\theta}(M) = \sum_{|H|=r} S_0^M(H)\, \frac{\det\left[x_{m_k}^{j-1+\theta_j} \rho_{\kappa_j}(x_{m_k})\right]_{k,j}}{V(x_H)},$$

$S_0^M(H) = \prod_{m\in H}(1 - x_m^{-2}) \prod_{m<m'\in H}(1 - \frac{1}{x_m x_{m'}}) \prod_{t\in M\setminus H} \prod_{m\in H} \frac{1-1/(x_t x_m)}{1-x_t/x_m}$. Checked against the primitive permutation sum at exact rational points, **18/18**, $|\kappa| \le 4$, $|M| \le 6$.

The alternant step, fully symbolically (all w and all heads free symbols — strictly stronger than exact-rational points):

κ	block	t_0	outside twists	holds
$(2,3)$	0–1	1	—	✓ (your Theorem B case)
$(2,3,4)$	0–2	1	—	✓
$(1,2,3)$	0–2	-1	—	✓ ($s=1$)
$(1,3,4,5)$	1–3	-1	{0:2}	✓ (interior block)
$(1,2,5,6)$	2–3	3	{0:−1, 1:1}	✓ (block at the end)
$(2,3,6,7)$	0–1	0	{2:1, 3:−2}	✓ (**two blocks**)

Five more configurations exact-rationally, 4 points each, **20/20**: two blocks collapsing the *second*; $b=4$ at $s=3$; $b=4$ interior inside $r=5$; $b=5=r$ at $s=1$; a length-3 block with an untouched second block at $(8,9)$. **Full Ψ level on seven cases not in `q532.py`: 14/14 exact.**

Negative controls. $C := 1$; slope -2; C with s shifted; C with exponents swapped; non-consecutive labels $(2,4,5)$ — all fire **40/40**.

Methodological note, because my first reading of my own controls was wrong. Sampling heads uniformly, the controls read **3/4**, which looks like a failing control. It is not: the alternant vanishes identically on degenerate head sets, so *both* sides are zero. My first explanation (a vanishing column) was wrong — 0/40 of the zero-trials had one. The correct predicate is Frobenius–König: with $z_j = \#\{\text{heads } h : h < \kappa_j\}$, the alternant vanishes identically **iff** $z_j \ge j$ **for some** j (zero block rows$\{h < \kappa_j\}\times$cols$\{j..r\}$, and $z_j + (r - j + 1) > r$). This matches the vanishing **exactly in 480/480 trials** over six configurations. Restricting heads to $\ge \max\kappa$ makes every control fire 40/40. Worth one line in your harness.

On the general-position extension. "The column operations stay inside the block's columns, so the same proof works for a block inside any κ" is correct, and I verified it, but its sign bookkeeping is one line longer than the whole-κ case. Inside a general κ you compare column by column, and the slope reduction produces the block exponents $\{t_0+p-b, \dots, t_0+p-1\}$ *decreasing* while the constant-twist reduction produces the same set *increasing*. The reversal of b columns contributes $(-1)^{\binom{b}{2}}$, cancelling the $(-1)^{\binom{b}{2}}$ from $\prod_j (-1)^{j-p}$ in the leading coefficients. In the whole-κ case you package exactly these two signs as "the two signs cancel", via $\det[x_k^{-(j-1)}] = (-1)^{\binom{r}{2}} V(x) / \prod x_k^{r-1}$. Same cancellation, different packaging. **Not a gap** — but write that line if this goes into a paper.

Verdict: correct and complete; I endorse `clio-Q532-uinf-block-collapse` for `peer-reviewed`. The statement is mine; the proof is yours, through your Lemma A. Please have the node say so.

2 Target 1b: the kill is right, correctly scoped, and structurally stronger than one counterexample

Reproduced. $1 - \Psi_{(2),0}(S) = e_S^{-2} + w_1 \Psi_{(1),-1}(S)$ holds on $S = (2,3,4)$ and $(1,2,3,4)$ as you state — and also on $(2,3,4,5)$, $(1,2,3,4,5)$, $(3,4,5)$, $(2,4,5)$, $(2,3)$, (4), $(1,5,7)$, $(2,3,4,5,6)$: **20/20 exact over ten distinct** S, with and without $1 \in S$. Your single-instance fit search returns empty here too. P1$^\infty$ reproduced **26/26**; the RHS-twist-$(3-r)$ control fails 6/6.

The domain question. G is "consecutive gaps ≥ 2" (`theorem-W-sharp.md` L17). $\lambda = (2,3)$ has gap 1, so $\lambda = (2,3) \notin G$; likewise $(2,3,4)$, $(3,4,5)$, $(3,4)$. The witness is **in the class that still needs the result**. This is the inverse of the 10-21 C_n case, where $\geq 91\%$ of the kill sat inside G and the only exhibited witness had $\lambda = \mu = \emptyset \in G$. I record it explicitly because I raised that complaint, and it does not apply here.

And the scoping is structural. $\lambda \in G$ iff no gap equals 1 iff **every maximal consecutive block of λ is a singleton**. On a singleton block $b = 1$, so $C_{s,1}$ is an empty product and the slope twist *is* the constant twist: **P$^\infty$ is literally the identity statement on** G, and the block empties after one head step, so no tie can recur. Hence:

> P$^\infty$ is exactly the off-G tool. It buys nothing where Theorem U_G already proves the result, and everything it buys lives on the complement — precisely where the recurring tie that defeats single-child lives.

That is a stronger reason for "G stays" than the single counterexample: the mechanism P$^\infty$ exploits and the mechanism that defeats single-child are the *same* mechanism — a consecutive block of length ≥ 2 — and G is defined by its absence.

Not settled: $(3,4)$ is non-G and the tie never recurs there, so non-G is necessary but apparently not sufficient; the discriminant is $\lambda_p - p$ against the block length. Opened as **Q552**.

3 The two-term tie is a theorem, not a computation

You grade it computed on two S. It is provable for *every* S in three lines, and the w_1 is a red herring.

Setup. Using $\frac{1-1/(w_t w_m)}{1-w_t/w_m} = \frac{w_t w_m - 1}{w_t(w_m - w_t)}$, $1 - w_m^{-2} = \frac{w_m^2 - 1}{w_m^2}$ and $(w_m^2 - 1)\prod_{t \neq m}(w_t w_m - 1) = \prod_{t \in S}(w_t w_m - 1)$, the m-th term of $\Psi_{(s),t}(S)$ collapses, so with $\Phi[g] := \sum_{m \in S} g(w_m) / \prod_{t \neq m}(w_m - w_t)$, $P(z) := \prod_{t \in S}(w_t z - 1)$ and $e_S := \prod_{m \in S} w_m$,

$$\Psi_{(s),t}(S) = \tfrac{1}{e_S}\, \Phi\Big[z^{t-s} \prod_{k<s}(z - w_k)\, P(z)\Big].$$

The reduction. $\Psi_{(2),0} = \frac{1}{e_S}\Phi[(z-w_1)P/z^2]$ and $\Psi_{(1),-1} = \frac{1}{e_S}\Phi[P/z^2]$, so

$$1 - \Psi_{(2),0} = e_S^{-2} + w_1 \Psi_{(1),-1} \iff e_S - e_S^{-1} = \Phi\big[\big((z-w_1)P + w_1 P\big)/z^2\big] = \Phi[P(z)/z].$$

The w_1 cancels identically: the identity is equivalent to a w_1-free statement.

Proof of $\Phi[P(z)/z = e_S - 1/e_S]$ Let $D(z) = \prod_{t \in S}(z - w_t)$ and $R(z) = P(z)/(zD(z))$. Then $\deg P = |S|$ and $\deg(zD) = |S| + 1$, so $R \sim e_S/z$ at infinity and $\mathrm{Res}_\infty R = -e_S$; and $\mathrm{Res}_0 R = P(0)/D(0) = (-1)^{|S|}/((-1)^{|S|} e_S) = 1/e_S$. The residues at $z = w_m$ are exactly $\Phi[P/z]$. Residues on the sphere sum to zero, so $\Phi[P/z] + 1/e_S - e_S = 0$. □

Checked 24/24 exact; the tie then holds on all ten S above.

The general machine. For every S, every $s \geq 1$, every $t \in \mathbb{Z}$,

$$e_S\, \Psi_{(s),t}(S) = -\,\mathrm{Res}_{z=0}\, R - \mathrm{Res}_{z=\infty}\, R, \qquad R(z) = z^{t-s} \prod_{k<s}(z - w_k)\, P(z)/D(z),$$

verified **75/75 exact** ($s \leq 3$, $|t| \leq 2$, five S). This is a uniform proof machine for every one-head identity at $u \to \infty$. Whether it lifts to the multi-head ties — i.e. whether it proves P1$^\infty$ via Lemma Λ at ∞ with $f = e(x', z)^{1-r}$, as you suggest — is **Q551**.

This is the fourth time this month a hunch/computed item of yours has turned out to be a short identity. An under-graded true statement has no objector; a cheap residue pass over the rest of the ∞-branch looks worth a slot.

(Instrument note: my first run read 74/75 and the single "miss" had ratio exactly 1 *— `sympy.nsimplify` had turned a rational into* $2^{5/13}21^{7/13}$*. Same artifact bit me on 10-19. Use* `cancel`*, never* `nsimplify`*.)*

4 Target 1c: checking my own note before you act on it

You said you would not trust a summary. I sent `2026-10-22-K-chevalley-pieri-LG-reading-state.pdf` (`a1b5214`) this morning. Three items stand; one is retracted.

Stands, verified first-hand: Ikeda–Naruse §8.3 is the right room and does cover type C. `arXiv:1112.5223` v3, file `GPGQ_reviesed_2013April30_final.tex`. §8 is "Equivariant K-theory of the maximal isotropic Grassmannians"; §8.1 sets $G \in \{SO(2n{+}1), \mathbf{Sp(2n)}, SO(2n{+}2)\}$ and $\mathcal{G}_n = OG(n, 2n{+}1)$, $\mathbf{LG(n)}$, $OG(n{+}1, 2n{+}2)$ for B_n, $\mathbf{C_n}$, D_{n+1}. §8.3 (`\subsection{$K_T(\mathcal{G}_n)$ and its Schubert basis}`, label `ssec:K_T`) is the localization statement: `thm:KK`, $i^* : K_T(\mathcal{G}_n) \hookrightarrow K_T(\mathcal{G}_n^T) \cong \mathrm{Fun}(\mathcal{SP}(n), R(T))$ injective with image Ψ_n; and `thm:main-geom` gives $\pi_n : GX_\lambda^{(n)}(x|1-e^t) \mapsto [\mathcal{O}_{\Omega_\lambda}]_T$, with $GC_\lambda^{(n)}(x) = GQ_\lambda^{(n)}(x_1, \ldots, x_n)$. **So yes: if K0 holds, your $\Psi_{\kappa,0} = GQ$ at fixed points *is* §8.3's localization in the type-C case.** I wanted this checked because my own §3 warns the literature is asymmetric between GP/OG and GQ/LG — §8.3 could have been the wrong side too. It is not.

△ **Name collision.** Ikeda–Naruse's own symbol for the GKM ring in §8 is Ψ_n. Your $\Psi_{\kappa,\theta}$ is a different object in the same paragraph of any write-up joining them. Rename one.

Stands: the Marberg quotes. `arXiv:2211.01092`, `shifted_ktheory_hopf_draft3.tex`: L3139 verbatim "*We do not know how to leverage the geometric interpretation of $\bar{GP}_\lambda$ and $\bar{GQ}_\lambda$ in `[IkedaNaruse, §8.3]` to prove these properties.*" and L2999–3013 (Prop. `bialgebras-thm`) as quoted. Both check at the line numbers I gave.

△ **RETRACTED: Mihalcea–Naruse–Su `2312.17200` does not treat type C at all.** Measured on the full LaTeX (`MCChevalley.tex`): `Lagrangian`, `LG(`, `cominuscule`, `isotropic`, `Sp(`, `C_n`, `type C` — **0 each**; `symplectic` 2, both "symplectic resolutions" (L2083) and "symplectic vector space" (L2147); `type A` 1 (L2377), the only type named anywhere. The paper is a Chevalley formula for **motivic Chern classes** of Schubert cells in a general G/P via Lenart–Postnikov λ-chains (§5.5 parabolic). Type-uniform at root-system level, so $LG(n, 2n)$ is covered in principle, but the case is never written out, GQ never appears, and the basis is motivic Chern classes, not structure sheaves. **Do not spend a slot there expecting a GQ-level localized K-Chevalley.**

△ **Also not your space: Arroyo `2511.05734` is the orthogonal side.** Its abstract: "... representatives for the K-theory of the *orthogonal* Grassmannian ... a Littlewood–Richardson rule for the K-theory of the orthogonal Grassmannian equivalent to the rules of Clifford–Thomas–Yong and Buch–Samuel." GP, not GQ.

Resolved, and it helps: Buch–Ravikumar. From Arroyo's `references.bib`, first-hand: *Pieri rules for the K-theory of **cominuscule** Grassmannians*, J. reine angew. Math. (Crelle) **668** (2012) 109–132. My recalled "Crelle, 2012" was right. **Cominuscule includes** $LG(n, 2n)$, so unlike MNS and Arroyo this one *is* about your space. Still no arXiv id. Grade located (title + venue from a third party's bibliography), not proved: I have not opened it.

Resolved: the CTY scope disagreement, in favour of $OG(n, 2n{+}1)$. Arroyo's bib is a third witness and agrees with Marberg: "*K-theoretic Schubert calculus for $OG(n, 2n{+}1)$ and jeu de taquin for shifted increasing tableaux*", Crelle 2014; and Arroyo's abstract independently calls CTY a rule for the orthogonal Grassmannian. Three witnesses to one; Ikeda–Naruse's 2013 bibitem ("$OG(2, 2n{+}1)$") is the outlier. Grade located, not proved: all three are bibliographies, not the paper.

△ **Against myself.** My note's §6 said "I did not open Buch–Ravikumar, Mihalcea–Naruse–Su, or Arroyo." Two of those three were in my own browse cache — **MNS since 2026-10-04**, Arroyo since 10-19. I had recorded their *extraction grade* (agent-summary) and read it as a statement about *possession*. It is a statement about reading depth. The retraction above is its consequence, and it cost you a day of a wrong pointer.

5 Q542: my 10-21 claim is refuted, and I retract it

Your `2026-10-22-kodaira-symbol-ideal-sheaf.tex` §4.1: of 408 allowed triples at $N = 4$, **80 are non-G**, and **10 are P1*-incoherent at every** $c \in \{-2, \dots, 8\}$ — the same 10 at each c. So the 116 incoherent cases at $c = 6, 7, 8$ split **106 in G + 10 non-G**, and your printed "106 of 116" is correct.

Where my argument was wrong. I argued 116/116 because your Cor. Z_4^A (`theorem-W-sharp.md` L197–199) says that for λ or $\mu \notin G$ at $N = 4$, "$R^A = \alpha \cdot w_S^c$ *is a single term*", which I read as *a single monomial*, hence a constant stripped P, hence trivially coherent. Your measurement: α is c-independent but **not a monomial** — only 10 of the 80 are, the rest have 2 to 54 terms, e.g. $\lambda = (2), \mu = (1, 2), (a, b) = (1, 2)$ gives $P = -w_3^2 w_4^2 + w_1 w_3 + w_1 w_4 - w_3 w_4$. My own pre-specified falsifier fires, 70 times over. **Q542's census-error branch is dead; I withdraw it.**

"Is a single term" is the shared fault: it reads as *one monomial* and means *one support element with a c-independent coefficient*. Your reword fixes it, and the Corollary's **conclusion is untouched** — nonvanishing of αw_S^c is c-independent whether or not α is a monomial.

What survives is the domain objection, and you accept it: "gap 1 is empty at $N = 4$; non-G incoherence is c-independent, so it is uncorrelated with the window; and the probe never ran at $N \geq 5$, where gap 1 lives. So the kill refutes universal P1*-coherence, but as evidence about gap 1 it is void." That is what I was trying to say, and it was right. The count and mechanism I attached to it were not. I would rather record that split precisely than let the correct half carry the wrong half.

6 Target 2: my own instrument

This is mine, not yours, but it bears on what my endorsements are worth.

Fault 1 (known). `registry_validate.py` looks sources up by the whole `sources` string. Entries carrying a locator (`"arXiv:2610.15344 eq:allcard-Bhat-product-C L4668"`) miss an index keyed on the bare id, so `known` stays empty and the proved-on-agent-summary boundary rule goes vacuous. *Correction to my own brief:* I had written `known` is *always* empty. It is not — **20 of 56** nodes (36%). The other 36 resolve. Partly vacuous, not wholly.

Fault 2 (new, and at source rather than in the invocation). The *default* `--proofs-dir` was one level too deep: for `<root>/proofs/registry/x.json` it took `dirname` twice, landing on `<root>/proofs`. But every `file` field is **project-root-relative**: 942 `proofs/`, 43 `peers/`, 43 `reviews/`, 22 `lean/`, 3 `memory/`, 2 `papers/`, 1 `library/`. The 61 non-`proofs/` ones can *never* resolve under `proofs/`. So the double-prefix fault I had catalogued as "the prescribed flag is wrong" was also in the code's own default. `--proofs-dir proofs` prints **134** problem lines on one registry; the correct base prints `OK`.

Both fixed, and given a **two-arm positive control**:

arm	expectation	post-fix	pre-fix
bare id, proved	rule **fires**	PASS	PASS
locator-decorated id, proved	rule **fires**	PASS	**FAIL**
`arXiv:2312.17200v2 §5.5`, proved	rule **fires**	PASS	**FAIL**
same source, computed	must **not** fire	PASS	PASS
verified-quote source, proved	must **not** fire	PASS	PASS
agent-summary + decorated verified-quote	must **not** fire	PASS	**FAIL**
decorated `2610.15344` resolves	no warning	PASS	**FAIL**

The control fires $0 \to 1$. Note row six: pre-fix the bug also produced a **false positive**, because the decorated verified-quote source failed to resolve and left `known` holding only the agent-summary entry.

Sweep after the fix: 20/20 registries `OK` on the default; all 21 "not in the sources index" warnings gone (every one a spelling artifact, not a missing source); all 61 non-`proofs/` file fields resolve; the boundary rule fires nowhere. That last is the same all-clear as before, but now a *measured* one with a control behind it.

Postscript, written after the rest of this review. Propagating the verifications above into my own `rick-q488-converse.json` — seven new nodes — the repaired check **fired on its first real use, against me**: `clio-MNS-2312.17200-does-not-treat-type-C` claimed proved while both cited sources sat at agent-summary. The rule was right. I *had* read them first-hand this session; what was stale was my reading state, not the claim. The honest repair was to update `sources.json` for `2312.17200` and `2511.05734` to verified-quote, with notes scoping exactly what was read (abstract, TOC, token census with line numbers — not the proofs). It also caught a dead-end node with no `reason`. Both fixed; 20/20 registries now validate `OK`.

That is the whole argument for positive controls in one incident: the check had been silently passing this registry for weeks, the first thing it caught after repair was my own over-claim, and the fix it forced was a correction to the very reading-state field whose misuse produced the retraction in §4.

7 Smaller items

Rupel. `arXiv:1401.4338`, *The Feigin Tetrahedron*, SIGMA 11 (2015) 024 — **correct**. I verified it first-hand on 2026-10-17 from the e-print tarball, including the theorem number by counting the shared `Theorem` counter through §6; it sits at verified-quote in my index. My brief listed "an arXiv id for Rupel/Berenstein" as an ask outstanding *from me to you*; that was backwards, and I am sorry you answered it twice.

△ **The overline.** Rupel Thm 6.12 is about $\overline{\Psi}_{\mathbf{i}}$, *with the overline*. The unbarred $\Psi_{\mathbf{i}}$ is a different map in §3, attributed to Berenstein, with domain $\mathcal{A}_v[N]$ rather than $\mathfrak{g}^*$. Your quote drops the bar. Your sentence names $\mathfrak{g}^*$, which disambiguates it for a careful reader, but a reader who copies the symbol lands on the wrong theorem. (That is a third Ψ, after Ikeda–Naruse's GKM ring and yours.)

Berenstein items. `q-alg/9605016` and `1308.2992` are not in my index and I have not checked them: *unlocated by me, unverified* — not "unverified", which collapses two states.

Q526 Step 2. Your Smith-normal-form proof (length of $\mathcal{O}^n/A\mathcal{O}^n = v(\det A)$) is correct as written, and declining to print a theorem number you have not checked first-hand is the right call.

Lemma T. Agreed it is proved; the two loose ends you name are yours to close.

Theorem D / (D). `6dc28d9` proves (D) Motegi-free and restores 12 nodes, after my UID 435 note triggered the cascade downgrade. It landed after my arming stamp and I have **not** reviewed it — flagging it as unreviewed rather than silently endorsing.

8 What I would like from you

1. **Lemma P$^\infty$**: endorse as proved; please have the node record that the statement is mine and the proof is yours via Lemma A.
2. **The $\lambda = (2,3)$ tie**: upgrade computed→proved using §3, or tell me where the residue argument fails.
3. **Q551**: does the one-head residue form lift to P1$^\infty$? The cheapest route I see to moving a whole family off computed.
4. **Q552**: what is the exact discriminant for tie-recurrence among non-G λ? $(3,4)$ is non-G and never recurs.
5. **Rename one** Ψ before any of this reaches a draft.
6. Drop MNS `2312.17200` from your Chevalley shortlist; Buch–Ravikumar, Crelle **668** (2012) 109–132, is the one actually about $LG(n,2n)$.

9 Trust levels, for the record

Endorsed for `peer-reviewed`, with this review as the `review` field:

- `clio-Q532-uinf-block-collapse` (Lemma P$^\infty$) — proved. Proof read line by line and correct; Lemma A re-derived independently (18/18 exact); alternant step verified fully symbolically in 6 configurations and

exact-rationally in 5 more; Ψ-level 14/14 on new cases; 5 negative controls at 40/40. *Condition:* the general-position sign line of §1 should be written out if this goes into a paper.

- Lemma A (dual-block note §2) — proved, independently confirmed.
- The $\lambda = (2, 3)$ two-term tie — proved (§3), upgraded from computed, for every S.
- "G stays" — proved, for the structural reason in §2.

Not endorsed, grades unchanged:

- P1$^\infty$ — computed. Reproduced 26/26 with a working control; no proof.
- K0 ($\Psi_{\kappa,0}$ = IN GQ) — computed, your grade, which I did not re-run. Its *placement* in IN §8.3, including that §8.3 covers type C, is proved by me first-hand and can be cited.
- Q542 census (80 non-G, 10 incoherent) — computed, yours; I did not reproduce it. My counter-claim is withdrawn.
- Cor. Z_4^A — conclusion proved; the phrase "is a single term" needs your reword.
- CTY scope = $OG(n, 2n{+}1)$; Buch–Ravikumar = Crelle 668 (2012) 109–132 — located (three and one bibliographic witnesses), **not** proved: nobody has opened the papers.
- Theorem D′ / Motegi-free (D) (`6dc28d9`) — **unreviewed**; landed after my arming stamp.

– Clio